

NT-C-013 | MEAN Stack | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Programming & Full Stack
Target learners	Learners seeking Angular and Node.js full-stack roles.
Prerequisites	Basic HTML, CSS and programming logic.
Approved duration	18-22 weeks / 220-280 guided hours.
Final outcome	Build and deploy a secure MongoDB, Express, Angular and Node.js application.
Assessment profile	Programming/Platform

Approved tools / platforms

TypeScript, Angular, RxJS, Node.js, Express, MongoDB, HTML, CSS, REST, Git.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Prefer runnable or demonstrable outputs, clear input validation, error handling, tests and version control.	Do not treat copied code or syntax recall as proof of implementation skill.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-013 | MEAN Stack | NT-C-013-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Language fundamentals and async programming; HTML, CSS and responsive layouts
Completion evidence	Typed exercises; Responsive pages
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 1 - JavaScript and TypeScript

Objective: Language fundamentals and async programming

Observable evidence: Typed exercises

Concept	Approved knowledge statement	Application behaviour	Evidence
Language fundamentals	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when language fundamentals is appropriate; apply it in a MEAN Stack task; verify the result.	Typed exercises
async programming	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when async programming is appropriate; apply it in a MEAN Stack task; verify the result.	Typed exercises

Module 2 - Web foundation

Objective: HTML, CSS and responsive layouts

Observable evidence: Responsive pages

Concept	Approved knowledge statement	Application behaviour	Evidence
HTML	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when html is appropriate; apply it in a MEAN Stack task; verify the result.	Responsive pages
CSS	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when css is appropriate; apply it in a MEAN Stack task; verify the result.	Responsive pages
responsive layouts	A layout that adapts to different screen sizes and input methods. In MEAN Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when responsive layouts is appropriate; apply it in a MEAN Stack task; verify the result.	Responsive pages

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Typed exercises; Responsive pages, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-013 | MEAN Stack | NT-C-013-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Components, templates, binding and directives; Routing, forms, RxJS and API calls
Completion evidence	Angular application; Multi-view SPA
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 3 - Angular core

Objective: Components, templates, binding and directives

Observable evidence: Angular application

Concept	Approved knowledge statement	Application behaviour	Evidence
Components	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when components is appropriate; apply it in a MEAN Stack task; verify the result.	Angular application
templates	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when templates is appropriate; apply it in a MEAN Stack task; verify the result.	Angular application
binding	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when binding is appropriate; apply it in a MEAN Stack task; verify the result.	Angular application
directives	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when directives is appropriate; apply it in a MEAN Stack task; verify the result.	Angular application

Module 4 - Angular services

Objective: Routing, forms, RxJS and API calls

Observable evidence: Multi-view SPA

Concept	Approved knowledge statement	Application behaviour	Evidence
Routing	Selection and forwarding of network paths between destinations. In MEAN Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when routing is appropriate; apply it in a MEAN Stack task; verify the result.	Multi-view SPA
forms	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when forms is appropriate; apply it in a MEAN Stack task; verify the result.	Multi-view SPA
RxJS	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when rxjs is appropriate; apply it in a MEAN Stack task; verify the result.	Multi-view SPA
API calls	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when api calls is appropriate; apply it in a MEAN Stack task; verify the result.	Multi-view SPA

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Angular application; Multi-view SPA, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-013 | MEAN Stack | NT-C-013-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Middleware, routing and REST; Modelling, CRUD, aggregation and indexes
Completion evidence	Backend service; Persistent data layer
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 5 - Node and Express

Objective: Middleware, routing and REST

Observable evidence: Backend service

Concept	Approved knowledge statement	Application behaviour	Evidence
Middleware	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when middleware is appropriate; apply it in a MEAN Stack task; verify the result.	Backend service
routing	Selection and forwarding of network paths between destinations. In MEAN Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when routing is appropriate; apply it in a MEAN Stack task; verify the result.	Backend service
REST	An http-oriented interface style using resources, methods and predictable responses. In MEAN Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when rest is appropriate; apply it in a MEAN Stack task; verify the result.	Backend service

Module 6 - MongoDB

Objective: Modelling, CRUD, aggregation and indexes

Observable evidence: Persistent data layer

Concept	Approved knowledge statement	Application behaviour	Evidence
Modelling	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when modelling is appropriate; apply it in a MEAN Stack task; verify the result.	Persistent data layer
CRUD	The create, read, update and delete lifecycle for persistent records. In MEAN Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when crud is appropriate; apply it in a MEAN Stack task; verify the result.	Persistent data layer
aggregation	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when aggregation is appropriate; apply it in a MEAN Stack task; verify the result.	Persistent data layer
indexes	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when indexes is appropriate; apply it in a MEAN Stack task; verify the result.	Persistent data layer

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Backend service; Persistent data layer, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-013 | MEAN Stack | NT-C-013-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Auth, validation, unit/API tests and Git; Build, environment, logging and documentation
Completion evidence	Secured tested solution; Deployed MEAN project
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 code/design scenarios (30) + 2 practical tasks (40)

Module 7 - Security and testing

Objective: Auth, validation, unit/API tests and Git

Observable evidence: Secured tested solution

Concept	Approved knowledge statement	Application behaviour	Evidence
Auth	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when auth is appropriate; apply it in a MEAN Stack task; verify the result.	Secured tested solution
validation	Checking input or output against defined rules before trusting it. In MEAN Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when validation is appropriate; apply it in a MEAN Stack task; verify the result.	Secured tested solution
unit/API tests	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when unit/api tests is appropriate; apply it in a MEAN Stack task; verify the result.	Secured tested solution
Git	Distributed version control for tracking and reviewing source changes. In MEAN Stack, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when git is appropriate; apply it in a MEAN Stack task; verify the result.	Secured tested solution

Module 8 - Deployment capstone

Objective: Build, environment, logging and documentation

Observable evidence: Deployed MEAN project

Concept	Approved knowledge statement	Application behaviour	Evidence
Build	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when build is appropriate; apply it in a MEAN Stack task; verify the result.	Deployed MEAN project
environment	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when environment is appropriate; apply it in a MEAN Stack task; verify the result.	Deployed MEAN project
logging	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when logging is appropriate; apply it in a MEAN Stack task; verify the result.	Deployed MEAN project
documentation	A defined competency within MEAN Stack. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when documentation is appropriate; apply it in a MEAN Stack task; verify the result.	Deployed MEAN project

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Secured tested solution; Deployed MEAN project, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-013 | MEAN Stack | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-013-SK01	JavaScript and TypeScript	Prerequisite foundation	Typed exercises	Critical
NT-C-013-SK02	Web foundation	M01 plus earlier foundations	Responsive pages	Critical
NT-C-013-SK03	Angular core	M02 plus earlier foundations	Angular application	Supporting
NT-C-013-SK04	Angular services	M03 plus earlier foundations	Multi-view SPA	Supporting
NT-C-013-SK05	Node and Express	M04 plus earlier foundations	Backend service	Supporting
NT-C-013-SK06	MongoDB	M05 plus earlier foundations	Persistent data layer	Critical
NT-C-013-SK07	Security and testing	M06 plus earlier foundations	Secured tested solution	Supporting
NT-C-013-SK08	Deployment capstone	M07 plus earlier foundations	Deployed MEAN project	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-013 | MEAN Stack | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-013-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of JavaScript and TypeScript and Web foundation?	The learner should provide Typed exercises; Responsive pages and explain how the work applies Language fundamentals and async programming; HTML, CSS and responsive layouts. Recall alone is insufficient.
NT-C-013-S01-EX02	Scenario/remediation	A learner reports completing Web foundation but cannot produce Responsive pages. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-013-S01, request Responsive pages, and allow a new equivalent test only after remediation.
NT-C-013-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Angular core and Angular services?	The learner should provide Angular application; Multi-view SPA and explain how the work applies Components, templates, binding and directives; Routing, forms, RxJS and API calls. Recall alone is insufficient.
NT-C-013-S02-EX02	Scenario/remediation	A learner reports completing Angular services but cannot produce Multi-view SPA. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-013-S02, request Multi-view SPA, and allow a new equivalent test only after remediation.
NT-C-013-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Node and Express and MongoDB?	The learner should provide Backend service; Persistent data layer and explain how the work applies Middleware, routing and REST; Modelling, CRUD, aggregation and indexes. Recall alone is insufficient.
NT-C-013-S03-EX02	Scenario/remediation	A learner reports completing MongoDB but cannot produce Persistent data layer. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-013-S03, request Persistent data layer, and allow a new equivalent test only after remediation.
NT-C-013-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Security and testing and Deployment capstone?	The learner should provide Secured tested solution; Deployed MEAN project and explain how the work applies Auth, validation, unit/API tests and Git; Build, environment, logging and documentation. Recall alone is insufficient.
NT-C-013-S04-EX02	Scenario/remediation	A learner reports completing Deployment capstone but cannot produce Deployed MEAN project. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-013-S04, request Deployed MEAN project, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Programming/Platform blueprint. Human-review every scored item before initial production release.

NT-C-013 | MEAN Stack | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Angular task app	Components, routing and forms	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Service-request API	Express, MongoDB and validation	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Operations portal	MEAN integration, security and deployment	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Prefer runnable or demonstrable outputs, clear input validation, error handling, tests and version control.

NT-C-013 | MEAN Stack | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Junior MEAN Developer	Angular 25% [critical]; Node 25% [critical]; MongoDB 25%; Git 25%	Deployed full-stack app	If a critical skill is below 60 or deployed full-stack app is missing, return Improve First or Needs Evidence with the linked module/project.
2	Angular Developer - Junior	Components 25% [critical]; RxJS 25% [critical]; forms 25%; APIs 25%	Angular SPA	If a critical skill is below 60 or angular spa is missing, return Improve First or Needs Evidence with the linked module/project.
3	Node.js Developer - Junior	Express 34% [critical]; REST 33% [critical]; data modelling 33%	Tested backend service	If a critical skill is below 60 or tested backend service is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-013 | MEAN Stack | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is MEAN Stack for?	Learners seeking Angular and Node.js full-stack roles.
What should I know before starting?	Basic HTML, CSS and programming logic.
How long is the approved learning journey?	18-22 weeks / 220-280 guided hours.
What will I be able to demonstrate?	Build and deploy a secure MongoDB, Express, Angular and Node.js application.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
MEAN Stack knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-013; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```